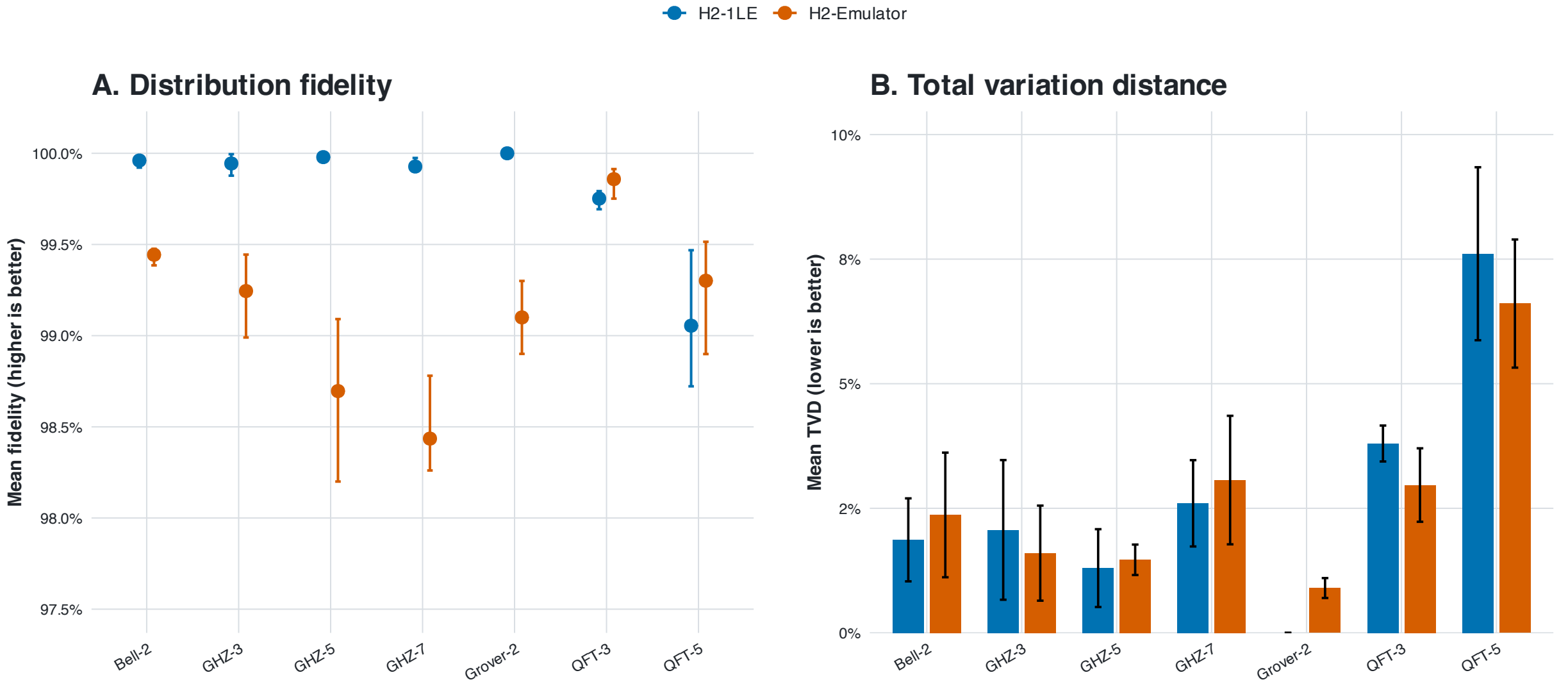


Quantinuum Nexus Full-Suite Emulator Validation

1,000 shots per circuit × 3 repetitions per target; error bars show observed range (A) or ±1 SD (B)



- QUESTION — Did the complete seven-circuit workflow return output distributions close to the exact ideal distributions?
- READ — Fidelity should be near 100%; TVD should be near 0%. Compare targets only within the same circuit.
- RESULT — Every target/circuit mean fidelity exceeded 98.4%; QFT-5 had the largest mean TVD on both targets.
- WHY IT MATTERS — The full workflow, circuit identity mapping, bit order, retrieval, and distribution scoring all worked end to end.
- BOUNDARY — These are emulator measurements, not physical trapped-ion QPU results; QFT measurement fidelity does not test the complete phase state.